Programming for Data Analytics 1

Portfolio of evidence (POE)

Student number:	**ST10543942**
Name and surname:	**Pfunzo Madzhiga**
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1. Dataset Justification

The Breast Cancer Wisconsin (Diagnostic) dataset is suitable for this classification task because it contains numerical measurements of cell nuclei obtained from breast mass images. The dataset has 569 observations, 30 predictive features, and a target indicating whether a tumour is benign or malignant. The dataset is appropriate because the target variable consists of two classes, making it a binary classification problem.

The numerical features can be used to identify patterns associated with the two classes. The dataset is also manageable for a beginner-level machine learning project and is available through Scikit-learn. Before modelling, the dataset will be checked for missing values, duplicate records, data types, and class distribution. These checks will help establish its suitability for analysis. As this is a diagnostic dataset, the model's predictions should be treated as educational results rather than as a substitute for professional medical diagnosis.

2. Algorithm Justification

Logistic Regression will be used because it is a supervised machine learning algorithm designed for classification problems. In this activity, it will classify observations as benign or malignant based on the numerical features in the dataset. The algorithm is suitable because it provides a relatively simple baseline model, is computationally efficient, and can estimate the probability of an observation belonging to a class.

Feature selection and standardisation will be used to improve the modelling process. Model performance will be evaluated using a confusion matrix, accuracy, precision, recall, F1-score, and ROC-AUC. Particular attention will be given to recall for the malignant class because failing to identify a malignant tumour is a serious classification error.

3. Analysis plan

Stage	Planned activity	Purpose
Exploratory Data Analysis	Inspect the dataset, data types, missing values, duplicates, and class distribution. Use graphs to explore the data.	Understand data quality and identify patterns.

Feature selection	Use the SelectKBest method with the ANOVA F-test to select the 15 most informative features from the training data.	Reduce the number of predictors and simplify the model.
Train model	Split the data into 80% training and 20% testing sets. Standardise features and train Logistic Regression with C=1.0 and max_iter=5000.	Build a baseline classification model.
Interpret and evaluate	Calculate accuracy, precision, recall, F1-score, and ROC-AUC. Examine the confusion matrix.	Assess performance, particularly the detection of malignant cases.
Write report	Document the dataset, data preparation, graphs, selected features, model settings, results, limitations, and recommendations.	Communicate the findings clearly.

4. Report

1. Introduction

The purpose of this study is to develop and evaluate a machine learning classification model that classifies breast tumours as benign or malignant. The study uses the Breast Cancer Wisconsin (Diagnostic) dataset and Logistic Regression to identify patterns in numerical measurements of cell nuclei.

The dataset contains measurements extracted from digitised images of breast mass samples and is suitable for demonstrating a binary classification problem. The study follows a structured process involving dataset selection, exploratory data analysis, data preprocessing, feature selection, model training, performance evaluation, and recommendations for improvement.

Machine learning classification can help identify patterns in diagnostic data. However, the model's predictions must be interpreted carefully and should not replace professional medical diagnosis.

2. Dataset Justification

2.1 Dataset source and suitability

The Breast Cancer Wisconsin (Diagnostic) dataset was developed by Wolberg, Mangasarian, Street, and Street and is available through the UCI Machine Learning Repository. It contains 569 observations and 30 numerical predictive features describing characteristics of cell nuclei, including radius, texture, perimeter, area, smoothness, and compactness. The dataset is suitable for this study because the target variable consists of two classes: benign and malignant. This makes it appropriate for supervised binary classification. The dataset is also available through the Scikit-learn Python library, which provides tools for implementing machine learning algorithms and evaluating their performance.

2.2 Data quality

The dataset was inspected using Python to examine its structure, data types, missing values, duplicate records, descriptive statistics, and target-class distribution. The UCI Machine Learning Repository reports that the original dataset has no missing values. These checks were important to verify the quality of the data before training the model.

The diagnosis variable was separated from the predictive features to prevent the model from using the answer as an input. The target was coded as 0 for benign and 1 for malignant to support evaluation of the model's ability to identify malignant cases.

2.3 Relevance and limitations

The dataset is relevant because its numerical measurements can be used to identify patterns associated with benign and malignant tumours. However, it has limitations. It is a specific diagnostic dataset and may not represent all patient populations or healthcare settings. The model also uses only the available numerical measurements and does not include every factor that may influence a clinical diagnosis. Therefore, successful classification on this dataset does not guarantee equivalent performance on new patients. The model should be considered an educational exercise rather than a clinically validated diagnostic tool.

3. Analysis Plan

The analysis was planned before model development to ensure a structured approach to data preparation, training, and evaluation.

Stage	Planned activity	Purpose

Exploratory Data Analysis	Inspect data structure, data types, missing values, duplicates, summary statistics, and class distribution. Create graphs.	Understand the data and identify quality issues and patterns.
Feature selection	Use SelectKBest with the ANOVA F-test to select 15 features.	Reduce the number of predictors.
Model preparation	Split the data into 80% training and 20% testing sets and standardise the selected features.	Prepare the data for modelling.
Model training	Train Logistic Regression with C = 1.0 and max_iter = 5000.	Establish a baseline model.
Model evaluation	Calculate accuracy, precision, recall, F1-score and ROC-AUC and examine the confusion matrix.	Assess classification performance and errors.
Model improvement	Compare C values of 0.01, 0.1, 1.0 and 10.0.	Determine whether alternative regularisation settings improve performance.
Reporting	Document methods, findings, limitations and recommendations.	Communicate results clearly.

4. Exploratory Data Analysis

4.1 Dataset structure and summary statistics

The dataset contains 569 observations and 30 predictive features. These features describe different characteristics of cell nuclei. Examples include mean radius, mean texture, mean perimeter, mean area and mean smoothness. Python functions such as head(), info(), and describe() were used to inspect the dataset, identify data types, and calculate descriptive statistics. These steps provided an initial understanding of the dataset before model training.

4.2 Missing values and duplicate records

The dataset was checked for missing values and duplicate records. The original UCI dataset is documented as having no missing values. The duplicate check was also performed to

identify repeated observations in the loaded data. These checks are important because data quality issues can affect model development and the interpretation of results. The notebook outputs were used to determine whether additional cleaning was required.

4.3 Target-class distribution

A count plot was used to examine the distribution of benign and malignant cases. The dataset contains 357 benign cases and 212 malignant cases (Scikit-learn, n.d.). The classes are therefore not equally represented. Stratified sampling was used when splitting the dataset to preserve approximately the same class proportions in the training and testing sets. Because the classes are not equally represented, accuracy alone may not provide a complete assessment of performance. Precision, recall, F1-score and ROC-AUC were also selected for evaluation (Scikit-learn, n.d.).

4.4 Relationships between features

A box plot was created to compare mean radius measurements between benign and malignant cases. This helped investigate whether the distribution of mean radius differed between the two diagnosis groups. A correlation heatmap was also used to examine relationships between selected numerical features. The heatmap showed the direction and strength of linear relationships between the variables. These visualisations helped identify patterns that may be useful for classification. However, correlation does not prove causation, and the usefulness of the features was further examined through feature selection and model evaluation.

5. Data Preprocessing and Feature Selection

5.1 Defining the features and target

The 30 numerical measurements were used as predictor variables, while diagnosis was used as the target variable. The target was coded as 0 for benign and 1 for malignant.

Separating the target from the predictors ensured that the model learned from the numerical measurements rather than using the diagnosis label as an input.

5.2 Training and testing sets

The dataset was divided into training and testing sets using an 80:20 split. The training set contained 455 observations, while the testing set contained 114 observations.

The training set was used to fit the model, while the testing set was reserved for evaluating predictions on observations that had not been used during training. A random state of 42 was

used to make the split reproducible. Stratified sampling was applied to maintain approximately the same class distribution in both sets.

5.3 Feature selection

SelectKBest with the ANOVA F-test was used to select 15 features from the original 30 predictors. SelectKBest ranks features according to the scores produced by the specified statistical test and retains the required number of features (Scikit-learn, n.d.).

Feature selection reduces the number of predictors and can simplify the model. However, the selected features should not automatically be interpreted as the only medically important features because their selection depends on the dataset and the method used. The feature selection process was fitted on the training data to avoid information from the testing set influencing model development.

5.4 Standardisation

StandardScaler was used to standardise the selected features before model training. Standardisation places the features on comparable scales by adjusting their means and standard deviations. This is useful for Logistic Regression because the scale of predictors can affect model optimisation and regularisation. The preprocessing steps were combined in a pipeline so that the scaling parameters were learned from the training data and then applied to the testing data.

6. Classification Algorithm and Model Training

6.1 Choice of Logistic Regression

Logistic Regression was selected because it is a supervised learning algorithm suitable for classification. It estimates the probability that an observation belongs to a particular class and uses a decision threshold to assign a predicted label. The algorithm is appropriate for this task because the target has two categories. Scikit-learn provides an implementation of Logistic Regression that can be integrated with feature selection and preprocessing tools (Pedregosa et al., 2011).

6.2 Model settings

The baseline model used the following settings:

Parameter	Setting	Purpose
Algorithm	Logistic Regression	Classifies observations into two categories.

Feature selection	SelectKBest	Selects 15 features using the ANOVA F-test.
Standardisation	StandardScaler	Places selected features on comparable scales.
C	1.0	Controls regularisation strength.
Maximum iterations	5000	Allows sufficient iterations for optimisation.
Training data	80%	Used to fit the model.
Testing data	20%	Used to evaluate the model.
Random state	42	Makes the data split reproducible.

The feature selection, standardisation and classification steps were combined in a pipeline. Scikit-learn supports pipelines to combine sequential data transformations and estimators into a consistent modelling process (Pedregosa et al., 2011).

7. Model Evaluation

7.1 Evaluation metrics

The model was evaluated using accuracy, precision, recall, F1-score, ROC-AUC and a confusion matrix. These metrics provide different perspectives on classification performance (Scikit-learn, n.d.).

Accuracy measures the proportion of all predictions that are correct. Precision measures the proportion of predicted malignant cases that are actually malignant. Recall measures the proportion of actual malignant cases correctly identified. The F1-score combines precision and recall, while ROC-AUC measures the model's ability to distinguish between the classes across different classification thresholds.

Recall for the malignant class is particularly important because a false negative occurs when a malignant case is incorrectly classified as benign. However, precision must also be considered because false positives may result in additional investigations.

7.2 Baseline model results

The baseline model was evaluated using the testing dataset. The actual results from the Jupyter Notebook should be presented below.

- Accuracy: 0.9737
- Precision: 1.0

- Recall: 0.9286
- F1-score: 0.963
- ROC-AUC: 0.9993

The metrics should be interpreted together rather than relying on accuracy alone. Accuracy indicates the overall proportion of correct predictions, while precision and recall provide more specific information about errors involving malignant classifications. ROC-AUC provides an additional measure of the model's ability to distinguish between the two classes (Scikit-learn, n.d.). The final interpretation should be based on the actual values produced by the notebook.

7.3 Confusion matrix

The confusion matrix compares actual diagnoses with predicted diagnoses. With benign coded as 0 and malignant coded as 1, it identifies true negatives, false positives, false negatives, and true positives. True negatives are benign cases correctly classified as benign. False positives are benign cases incorrectly classified as malignant. False negatives are malignant cases incorrectly classified as benign, while true positives are malignant cases correctly identified.

The confusion matrix should be examined to determine the types of errors made by the model. Particular attention should be given to false negatives because failing to identify a malignant case is a serious limitation in a medical classification context. The confusion matrix generated in the notebook should be included in this section and discussed using its actual values.

7.4 Model improvement

To investigate possible improvements, different regularisation settings were compared using C values of 0.01, 0.1, 1.0 and 10.0. In Logistic Regression, smaller C values apply stronger regularisation, while larger C values apply weaker regularisation. Each model was assessed using the same training and testing split and the selected evaluation metrics. This comparison helped determine whether alternative regularisation settings produced a more suitable balance between precision, recall, and overall classification performance.

The preferred setting should be selected using the actual comparison results. Since identifying malignant cases is important, recall should receive particular attention, alongside precision, F1-score, and ROC-AUC. If the alternative settings do not improve performance meaningfully, the baseline model should be retained and its limitations discussed.

8. Discussion of Findings

The analysis demonstrates how a breast cancer dataset can be prepared and used to train a binary classification model. Exploratory data analysis provided information about the dataset's structure, the distribution of diagnoses and relationships between selected features. Feature selection reduced the number of predictors from 30 to 15, while standardisation prepared the selected features for Logistic Regression. The training and testing procedure allowed performance to be assessed using observations that were not used to fit the model.

The evaluation metrics and confusion matrix provide complementary information about performance. Accuracy summarises the proportion of correct predictions, while precision and recall help explain the balance between false positives and false negatives. ROC-AUC provides further information about the model's ability to distinguish between benign and malignant cases (Scikit-learn, n.d.).

The final assessment must be based on the actual notebook results. Strong performance on this dataset would indicate that the selected features contain useful information for distinguishing between the two classes. However, good performance on one dataset does not establish that the model will generalise to new patients or different clinical settings.

9. Limitations

The study has several limitations.

First, the dataset is relatively small and may not represent all patient populations or healthcare settings. Second, the model uses numerical measurements of cell nuclei and does not include every factor that may be relevant to a clinical diagnosis. Third, Logistic Regression may not fully capture complex nonlinear relationships between predictors and the target.

Fourth, selecting 15 features using an ANOVA F-test may exclude features that are useful through interactions or nonlinear relationships. Finally, the model was evaluated using one train-test split. Performance may differ when a different sample is used. Further validation would therefore be necessary before considering any real-world application.

10. Recommendations

The following recommendations are proposed:

- Prioritise malignant-case recall. Examine the model's ability to identify malignant cases and investigate any false negatives, while also considering precision and false positives.

- Use cross-validation. Apply stratified cross-validation to the training data to obtain a more reliable estimate of model performance and compare parameter settings.
- Compare alternative algorithms. Evaluate Logistic Regression against other classification methods, such as Support Vector Machines and Random Forests, using consistent validation procedures.
- Review feature selection. Investigate whether selecting a different number of features improves performance. Feature selection should remain within the training and validation process to avoid data leakage.
- Assess generalisability. Test the model on independent data from different sources or populations before considering practical use.
- Maintain human oversight. Treat the model as an educational and analytical tool rather than a replacement for professional medical assessment.

These recommendations are intended to improve the reliability of the modelling process and encourage responsible interpretation of classification results.

11. Conclusion

This study applied Logistic Regression to the Breast Cancer Wisconsin (Diagnostic) dataset to classify tumours as benign or malignant. The process included data exploration, preprocessing, feature selection, model training and evaluation using several classification metrics.

The approach demonstrated the importance of understanding data quality, selecting suitable predictors, and evaluating a model on observations not used during training. Comparing regularisation settings also provided a structured way to investigate possible improvements.

The conclusion regarding the model's effectiveness must be based on the actual notebook results, particularly its ability to identify malignant cases. Although the dataset is suitable for an educational classification exercise, the model would require further validation and testing before it could be considered for clinical use.

References

Cortes, C. and Vapnik, V. (1995) ‘Support-vector networks’, *Machine Learning*, 20, pp. 273–297. Available at: https://doi.org/10.1007/BF00994018.

Pedregosa, F. et al. (2011) 'Scikit-learn: Machine learning in Python', *Journal of Machine Learning Research*, 12, pp. 2825–2830. Available at: https://jmlr.org/papers/v12/pedregosa11a.html.

Scikit-learn (n.d.) *Breast cancer Wisconsin (diagnostic) dataset*. Available at: https://scikit-learn.org/stable/modules/generated/sklearn.datasets.load_breast_cancer.html

Scikit-learn (n.d.) *LogisticRegression*. Available at: https://scikit-learn.org/stable/modules/generated/sklearn.linear_model.LogisticRegression.html